

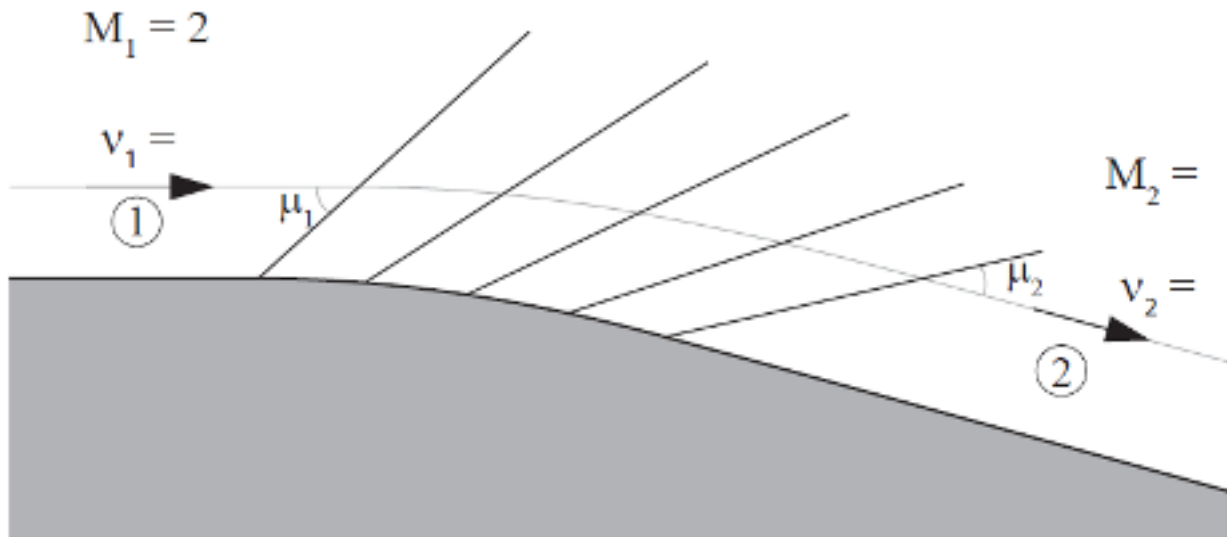
AERO50001 Aerodynamics 2 Problem Set #5

Extra practice: Expansion fans, shock-expansion theory

5.1 Expansion fan

★★☆ 15-25 mins

Determine the following quantities downstream of the continuous 10° convex corner shown below, with the flow upstream of the corner being at Mach 2:



- (a) The Mach number, M_2 .
- (b) The pressure, p_2 , as a function of the upstream pressure p_1 .
- (c) Find the angles μ_1 and μ_2 that the forward and rearward Mach lines form with respect to their local flow.

(a) Left-running : $\theta_1 + \gamma(M_1) = \theta_2 + \gamma(M_2)$

$$\therefore \gamma(M_2) = \theta_1 + \gamma(M_1) - \theta_2 = 0 + 26.38^\circ - (-10^\circ) = 36.38^\circ$$

$$\therefore M_2 = 2.39$$

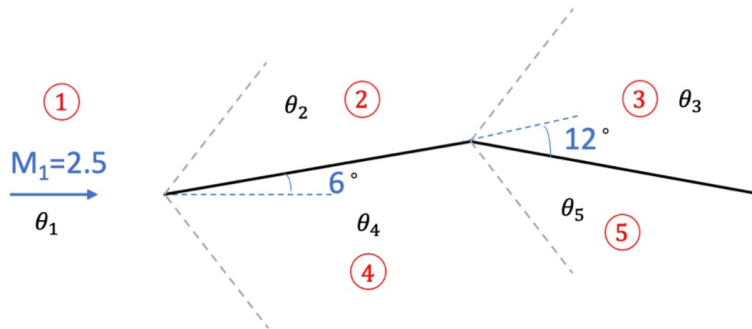
(b) $\frac{p_2}{p_1} = \frac{p_2}{p_{a2}} \frac{p_{a2}}{p_{a1}} \frac{p_{a1}}{p_1} = \frac{p_2}{p_{a2}} \frac{p_{a1}}{p_1} = \frac{7.8244}{14.393} = 0.5436$

(c) $\mu_1 = 30^\circ$, $\mu_2 = 24.73^\circ$

5.2 Cranked aerofoil

★★☆ 15-30 mins

Use shock-expansion theory to find the pressure on every surface of the following cranked aerofoil in a Mach 2.5 flow (relative to p_1).



At ②, $M_2 = 2.251$

$$\therefore \frac{p_2}{p_1} = 1.468$$

At ④, $\theta_1 - \nu(M_1) = \theta_2 - \nu(M_4)$

$$\therefore \nu(M_4) = \theta_2 - \theta_1 + \nu(M_1) = 6^\circ - 0^\circ + 39.12^\circ = 45.12^\circ$$

$$\therefore M_4 = 2.77, \frac{p_4}{p_1} = \frac{p_4}{p_{o4}} \frac{p_{o4}}{p_{o1}} \frac{p_{o1}}{p_1} = \frac{17.0859}{25.922} = 0.6591$$

At ③, $\theta_2 + \nu(M_2) = \theta_3 + \nu(M_3)$

$$\therefore \nu(M_3) = \theta_2 + \nu(M_2) - \theta_3 = 6^\circ + 33.02^\circ - (-6^\circ) = 45.02^\circ$$

$$\therefore M_3 = 2.77, \frac{p_3}{p_1} = \frac{p_3}{p_{o3}} \frac{p_{o3}}{p_{o2}} \frac{p_{o2}}{p_2} \frac{p_2}{p_1} = \frac{11.5812}{25.922} \cdot 1.468 = 0.4193$$

At ⑤, $\theta_4 - \nu(M_4) = \theta_5 - \nu(M_5)$ ($\theta_{\text{local}} = 12^\circ$)

$$\therefore \nu(M_5) = \theta_3 - \theta_4 + \nu(M_4) = -6^\circ - 6^\circ + 45.12^\circ = 33.12^\circ$$

$$\therefore M_5 = 2.2275, \frac{p_5}{p_1} = \frac{p_5}{p_4} \frac{p_4}{p_1} = 2.223 \cdot 0.6591 = 1.465$$